CS & IT ENGINEERING



Algorithms

Divide & Conquer



Lecture No.- 06

By- Aditya sir

Recap of Previous Lecture



- Topic
- Topic

QuickSort
↳ Partition Algo
↳ TC, SC.

Topics to be Covered



- Topic
- Topic
- Topic

Imp Questions on QS.

Matrix Multiplication



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.

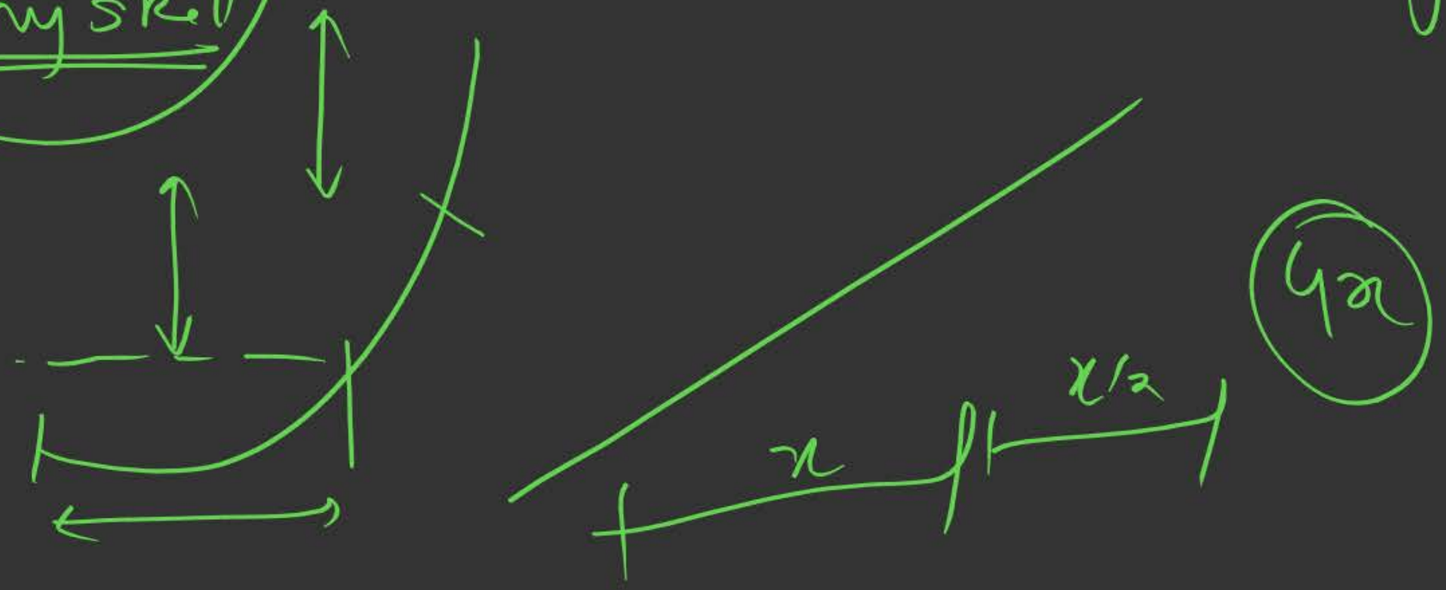


Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

Saturday
(Tomorrow 17th May) ⟶ 8 AM

2hr + 30 min
(pending)

Any skill





Topic : Divide & Conquer

Quick Sort :

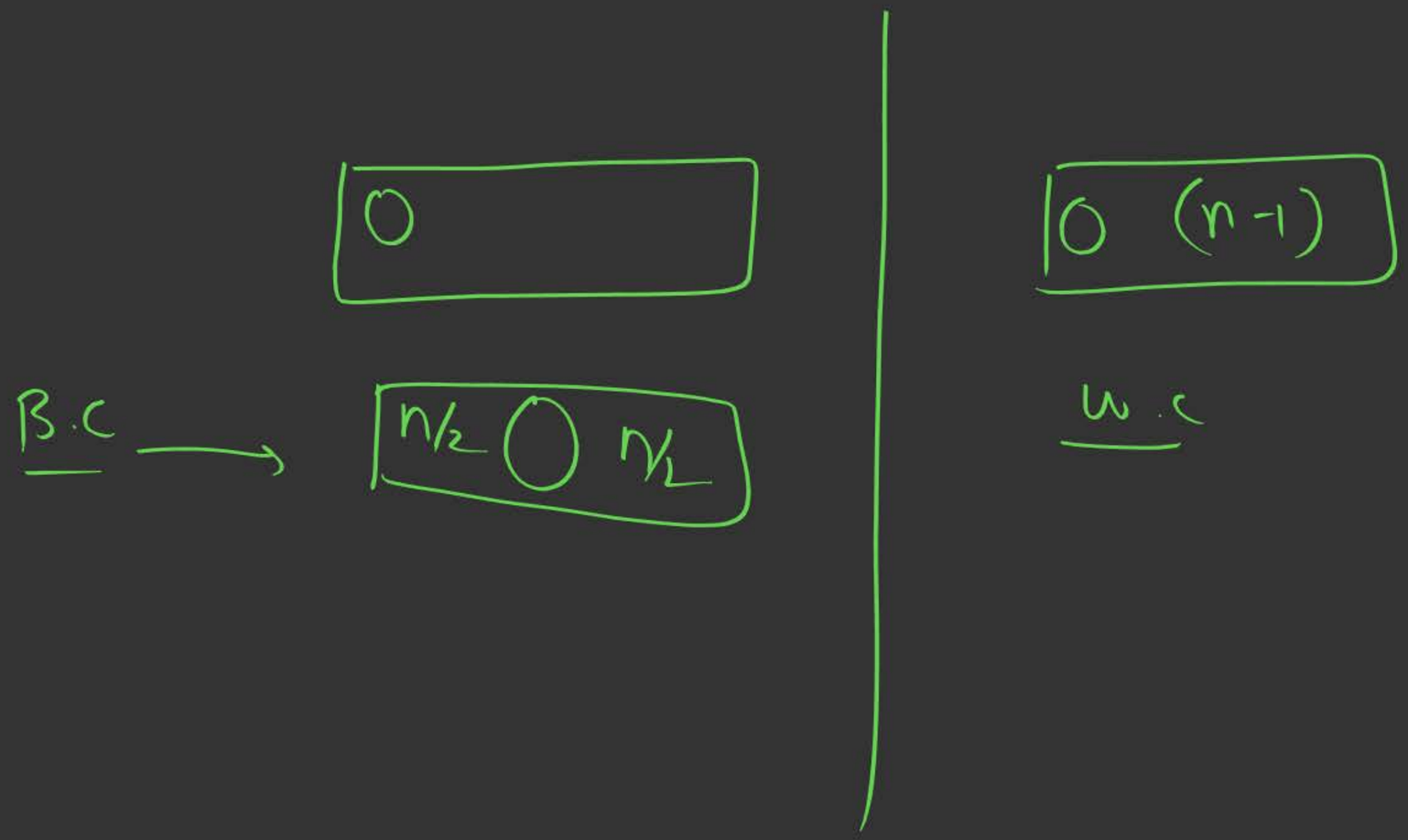
1. Time Complexity :

(1) Best case : Avg $O(n\log n)$: Unsorted input → (random)



(2) Worst case : $O(n^2)$: Sorted input → (asc/disc)




O
O (n-1)
B.C
n/2 O n/2
W.C



Topic : Divide & Conquer

Space Complexity :

1. Best Case : $O(\log n)$ → Recursion stack
2. Worst Case : $O(n)$ → ~~Spaced~~ Stowed Recursion stack





Topic : Divide & Conquer

#Q. Let P be a quick sort program to sort numbers in ascending order. Let t_1 and t_2 be the time taken by the program for the inputs [1 2 3 4] and [5 4 3 2 1] respectively. Which of the following holds?

44%

A $t_1 = t_2$ → 40%

B $t_1 > t_2$

C $t_1 < t_2$

D $t_1 = t_2 = = 5 \log 5$

Soln :- $t_1 \rightarrow i/p_1 \rightarrow$ asc $\quad n_1 = 4$

$t_2 \rightarrow i/p_2 \rightarrow$ desc $\quad n_2 = 5$

→ Asymptotically both are $O(n^2)$

→ Compare Absolute

$n_1 = 4$

$n_2 = 5$ → $t_1 < t_2$



Topic : Divide & Conquer

#Q. Let P be a Quick Sort Program to sort numbers in ascending order using the first element as pivot. Let t_1 and t_2 be the number of comparisons made by P for the inputs {1, 2, 3, 4, 5} and {4, 1, 5, 3, 2} respectively. Which one of the following holds.

62:1.

A $t_1 = 5$

B $t_1 < t_2$

C $t_1 > t_2$

D $t_1 = t_2$

asc

Soln t_1 i/p = {1,2,3,4,5} → n1 = 5

t_2 i/p = {4,1,5,3,2} → n2 = 5

↳ random

$t_1 > t_2$



Topic : Divide & Conquer

#Q. Quick Sort is run on two inputs shown below to sort in ascending order taking first element as pivot

i. 1, 2, 3 . . . . N

ii. n, n – 1, n – 2, . . . , 2, 1

Let C_1 and C_2 be the number of comparisons made for the inputs (i) and (ii) respectively. Then,

A $C_1 > C_2$

B $C_1 < C_2$

C $C_1 = C_2$

D We can't say anything for arbitrary n

i/p1 → 1, 2, 3 . . . n → asc $n_1 = (n)$

i/p2 → n, n-1 . . . . 1 → desc $n_2 = (n)$

→ W.C

$C_1 = C_2$ as $n_1 = n_2$



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V. Imp type

#Q. Assume that merge sort takes 30 sec to sort 64 elements in worst case. What is the approximate number of elements that can be sorted in the Worst case using merge sort using 6 minutes?

NAT

8.75 /

Soln :- Given $t = 30$ sec, $n = 64$ } wc

$t = 6$ mins, $n = ?$ } wc

wc, n, $TC = O(n \log_2 n)$

$a \approx n$

$a = c * n$

given

$t_1 = c * n \log_2 n$ ☆

$n = 64$, $t = 30$ sec.

$30 = c * 64 \log_2 64$

$30 = c \times 64 \times 6$

$$c = \frac{30}{64 \times 6}$$

how, in 6 mins

$t = 6$ min

$= 6 \times 60$ sec

$t = C * n \log n$

$6 \times 60 = C \times n \log_2 n$

$6 \times 60 = \frac{30}{64 \times 6} \times n \log_2 n$

$n \log_2 n = \frac{6 \times 60 \times 64 \times 6}{30} \Rightarrow$

$n \log_2 n = 6 \times 2 \times 6 \times 64$

Let $n = 2^k$

$2^k \times \log_2(2^k) = 6 \times 2 \times 6 \times 64$

$K \times 2^K = 6 \times 2 \times 64 \times 6$

$K \times 2^K = 4608$

$\underline{K=8}$ $\quad 8 \times 2^8$

$= 8 \times 256$

$= \underline{2048}$ ✗

$K=9$, $\quad 9 \times 2^9$

9×512 ✓

$= \boxed{4608}$

$n = 2^K$

$n = 2^9$

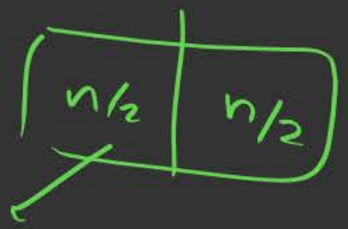
$= \boxed{512}$ ✓

$$\begin{array}{r} 64 \\ 72 \\ \hline 128 \\ 448\ \times \\ \hline 4608 \end{array}$$

Bonus

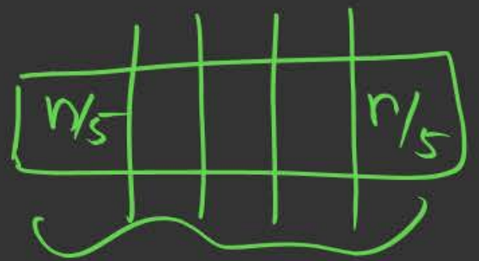
Conceptual

MS → $O(n \log_2 n)$



AJ_MS

TC = $O(n \log_5 n)$



each n/5

$n \rightarrow n/5 \rightarrow n/5^2 \rightarrow n/5^3 \cdots 1$

$n = 5^k \rightarrow k = \log_5 n$



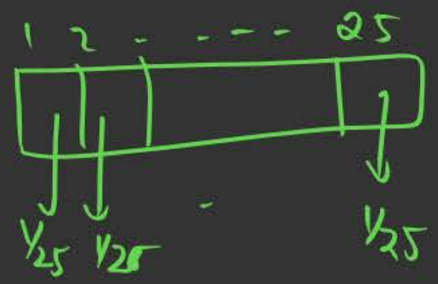
Topic : Divide & Conquer

#Q. An array of 25 distinct elements is to be sorted using quicksort. Assume that the pivot element is chosen uniformly at random. The probability that the pivot element gets placed in the worst possible location in the first round of partitioning (rounded off to 2 decimal places) is _____.

67%

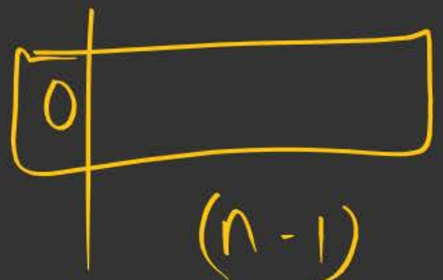
ans * 100 → int

Soln :- $n = 25$



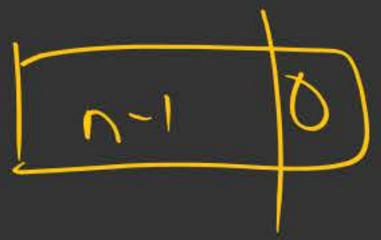
$P(\text{pivot} = \text{at a location}) = \frac{1}{25}$





Way 1 $\Rightarrow P(WC) = \frac{2}{25} = 0.08$

OR



Ways:-

$$P(wc) = P(1^{st}\ loc) + P(last\ loc)$$

$$= \frac{1}{25} + \frac{1}{25}$$

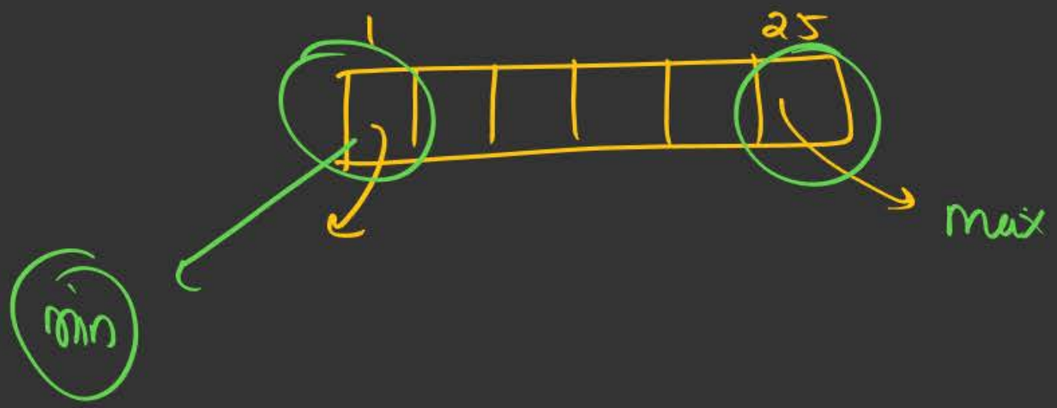
$$= \frac{2}{25}$$

$$= 0.08$$

$0.08 * 100$

$= 8$ ✓

Query 3 :-



$$P(wc) = P(\text{pivot} = \min \text{elem}) + P(\text{pivot} = \max \text{elem})$$

$$= \frac{1}{25} + \frac{1}{25}$$

$$= \frac{2}{25} = 0.08$$

1) Min-max
2) B.S
3) M.S
4) Q.S



Topic : Divide & Conquer

Matrix Multiplication Problem

w.r.t. square matrix of size n × n.



Topic : Divide & Conquer

Basics :

$$A_{n\times n} * B_{n\times n} \rightarrow C_{n\times n}$$

In general : $A_{r1\times c1} * B_{r2\times c2} = C_{r1\times c2}$

A and B are multiplicable only if $\boxed{c_1 = r_2}$



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Eq : Matrix Addition

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

TC

$$A + B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2\times2} + \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}_{2\times2} = \begin{bmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}_{2\times2}$$

$$A_{n\times n} + B_{n\times n} \rightarrow C_{n\times n}$$



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Time Complexity of Previous Method of Matrix Addition

$A_{n\times n} + B_{n\times n} \Rightarrow$

```
for i : 1 → n
        for j : 1 → n
                C[i, j] = A[i, j] + B[i, j]
```

TC ⇒ $\boxed{O(n^2)}$ [Matrix Addition]

and

Matrix subtraction



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Matrix Multiplication :

Given two square matrices each → $n \times n$

$$\underset{A}{\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}} * \underset{B}{\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}} = \begin{bmatrix} 1\times5+2\times7 & 1\times6+2\times8 \\ 3\times5+4\times7 & 3\times6+4\times8 \end{bmatrix}_{2\times2}$$



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non D&C

Matrix multiplication in general : (~~Non D and C~~)

A × B

2×2 2×2

$$\underset{A}{\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}} * \underset{B}{\begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}} = \underset{C}{\begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}}$$

Then, $t_1 \rightarrow$

		Add	mul
$c_{11} = a_{11} * b_{11} + a_{12} * b_{21}$	→	1	2
$c_{12} = a_{11} \times b_{12} + a_{12} \times b_{22}$	→	1,	2
$c_{21} = a_{21} \times b_{11} + a_{22} \times b_{21}$	→	1,	2
$c_{22} = a_{21} \times b_{12} + a_{22} \times b_{22}$	→	1,	2

Total

4 Addition & 8 Multi.



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Code of prev matrix multiplication logic (Non DnC)

$A_{n\times n} + B_{n\times n} = C_{n\times n}$

```
For i : 1 → n :
    For j : 1 → n :
        For k : 1 → n
            C[i, j] += A[i, k] * B[k, j]
```

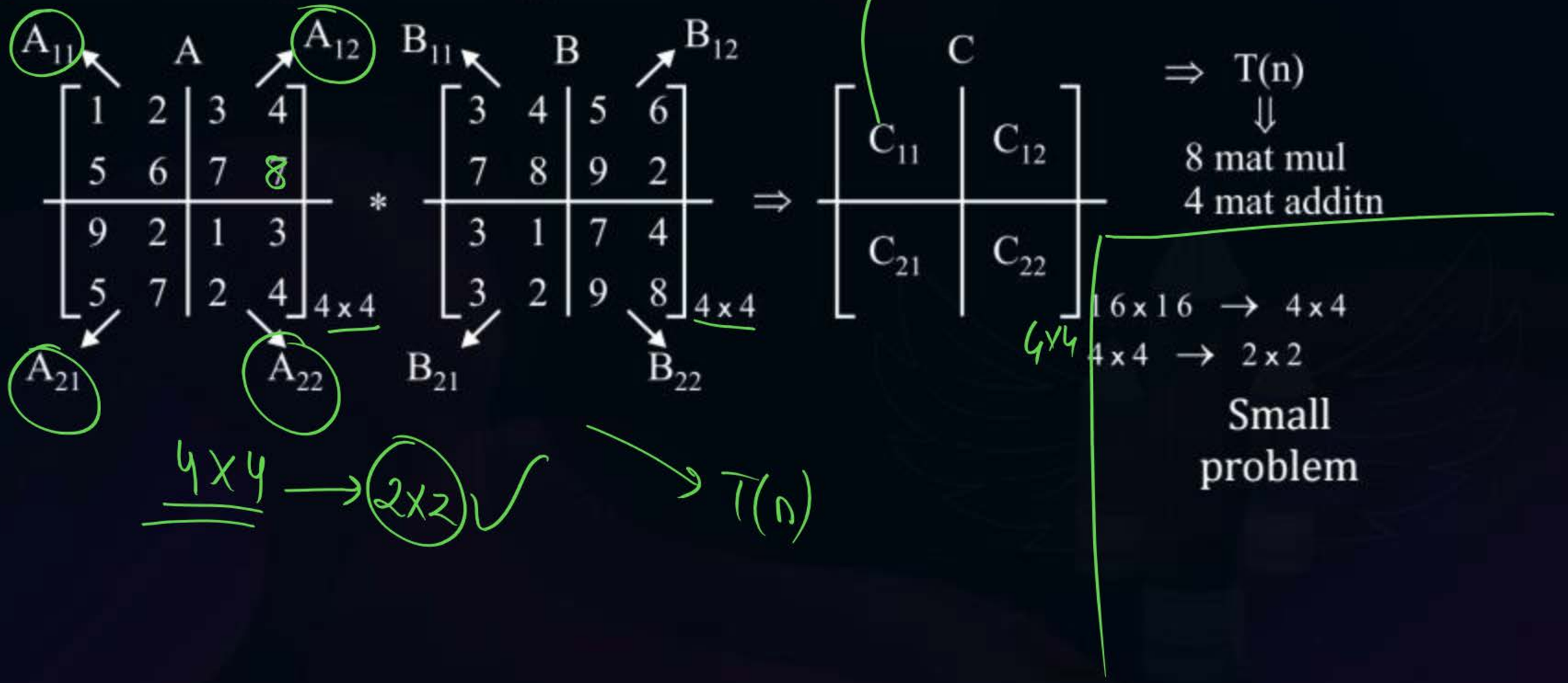
⇒ O(1) space

⇒ $TC = O(n^3)$

Can we do better using DnC approach >

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Eg : Divide & Conquest based approach





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Sub-problem

Where,

$$C_{11} = A_{11} * B_{11} + A_{12} * B_{21} - (1)$$
$$C_{12} = A_{11} * B_{12} + A_{12} * B_{22} - (2)$$
$$C_{21} = A_{21} * B_{11} + A_{22} * B_{21} - (3)$$
$$C_{22} = A_{21} * B_{12} + A_{22} \times B_{22} - (4)$$

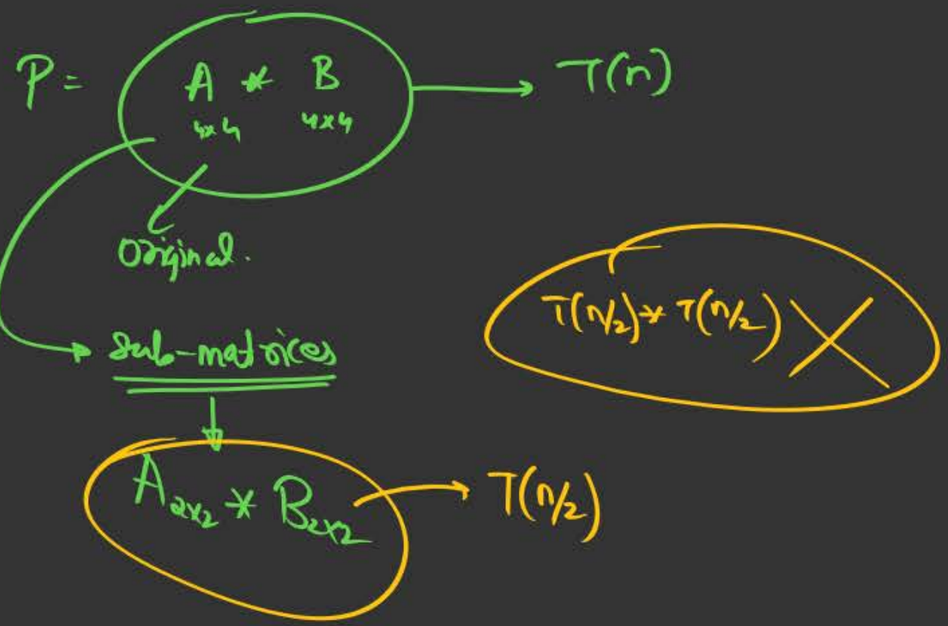
$A, B \rightarrow n \times n$

Mat Add	Mat Mul
1,	2
1,	2
1,	2
1,	2

4 sub-matrix addition & 8 sub-matrix Mul

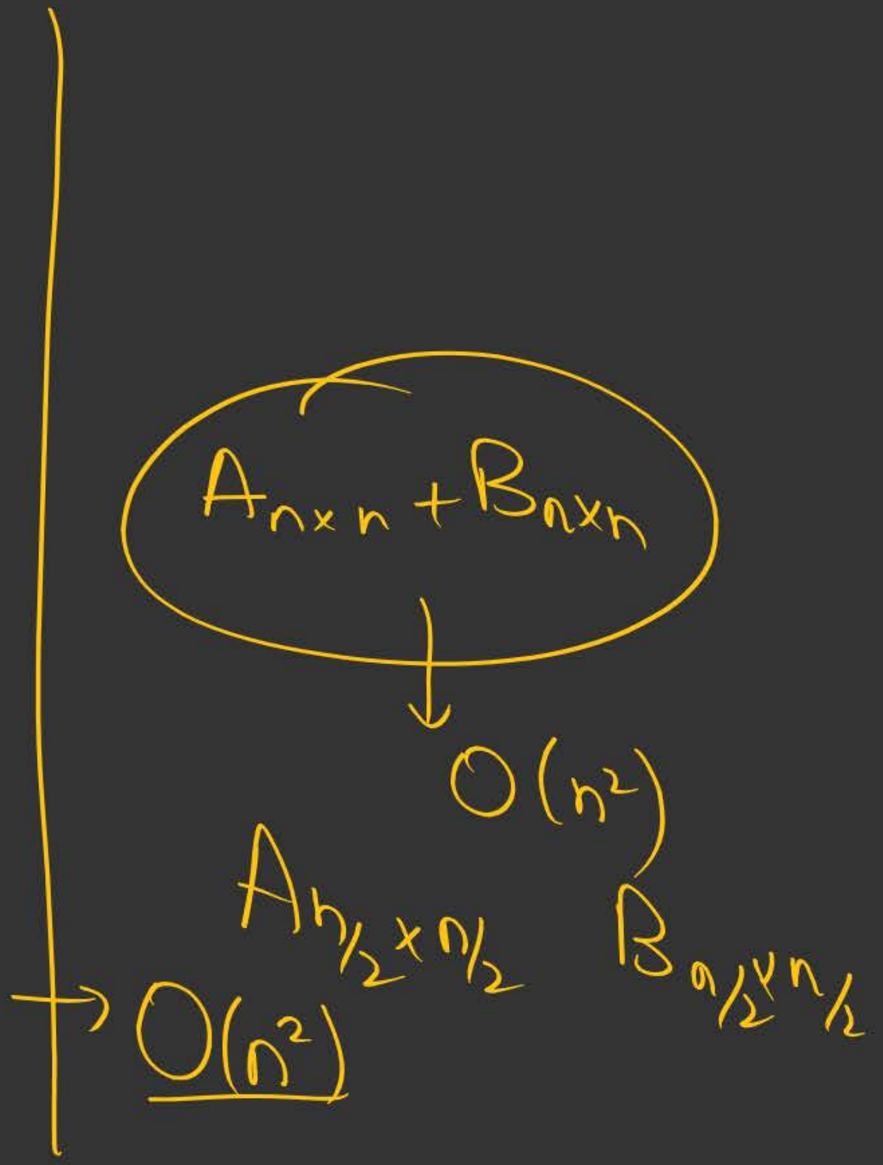
Combine

Subproblem → Divide



$$T(n) = 8 \cdot T(n/2) + 4 * O(n^2)$$

→ gate





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Divide :

$n \times n \rightarrow \left(\frac{n}{2} \times \frac{n}{2}\right)$ Subproblem

↓ Original problem

↓ $n \times n$

↓ $\frac{n}{2} \times \frac{n}{2}$

H.W

$T(n) = 8 * T(n/2) + O(n^2)$ → gate

H.W

Solve Recurrence using Back Sub

$$T(n) = 8T(n/2) + O(n^2)$$

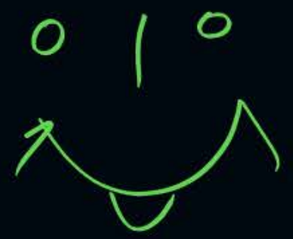


2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

Q S

Matrix Mul



THANK - YOU